Representation-Dependent Harmonic Alignment in the UNESCO World Heritage Corpus: Domed Monuments, Cluster Asymmetry, and Coordinate Choice

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May 4, 2026

Abstract

We analyze UNESCO Cultural and Mixed World Heritage sites ($N = 1011$) with a harmonic-alignment scoring framework. A global sweep over frequency and phase selects a frame near 3° spacing and the ∼35°E corridor. Subsequent analyses describe longitude deviations within that selected frame. Weak but reproducible structure appears across architectural subsets and related Eurasian datasets. The pattern is not recovered by invariant spatial methods and does not appear as a spectral periodicity. The observed alignment depends on the chosen representation and does not support a periodic generative model or coordinated placement system. The study treats this as a methodological case: clustered cultural datasets can yield stable template alignments that are not invariant spatial periodicities. The pipeline is fully reproducible.

Keywords: template alignment, spatial statistics, Levant, UNESCO World Heritage, 3-degree harmonics, representation dependence, circular statistics, monument distribution, Mount Gerizim

1 Introduction

This paper examines how a coordinate-dependent scoring framework can produce reproducible phase-localized alignment patterns in a geographically clustered corpus. The corpus is UNESCO Cultural and Mixed World Heritage sites ($N = 1011$). A global parameter sweep over frequency and phase selects a high-scoring frame near 3° spacing and the ∼35°E corridor (§4.1, §5.2). All subsequent analyses are conditional on this selected frame. Because the frame is chosen by optimizing over the data, results at this frame reflect post-selection and should not be read as independent statistical evidence. Gerizim provides one historically documented point within the identified corridor (§2.2). Within this frame, the data show weak but reproducible departures from spatial uniformity. The same pattern is absent

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at non-optimal anchors and is not recovered as a spectral peak under invariant periodogram analysis.

Three diagnostic checks point in the same direction: dome von Mises mixture, Spearman cluster asymmetry, and phase-rotation null (Table 3; $p_{\text{adj}} = 0.004, 0.002, 0.047$). Effects are small (A+ rate 13.1% vs. 10% null; $\rho = -0.1076$); the anomaly localizes the ∼35°E corridor, not a unique anchor. This is compatible with 3° spacing but does not establish it (§2.2, §4.1).

The observed spatial regularity is concentrated in the longitude range where Eurasian cultural site density is highest (the Eurasian high-density corridor, often referred to as the Silk Road region) and is consistent with the geographic clustering of culturally related sites. The Wikidata Q180987 stupa inventory and OWTRAD route network show similar enrichment patterns under the same reference frame, but share geographic and cultural structure with UNESCO. Their agreement is cross-corpus consistency, not independent confirmation (§7.3).

2 Background

2.1 Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history (Thom, 1967; Kendall, 1974; Freeman, 1976). Probability-based frameworks for archaeoastronomical alignments have been advanced by Magli (Magli, 2013, 2016) and Ruggles and Silva (Ruggles, 1999, 2015; Silva, 2020); the ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for World Heritage analysis.

2.2 Reference Phase and Candidate Anchors

The global anchor sweep (§5.2) places the ∼35°E corridor among the highest-scoring reference phases under the harmonic-deviation score, with marginal discovery-corrected support: both Gerizim and Jerusalem place the 35.25°E modulo-3° phase band in the global top 2.3% of 36,000 trial anchors, while Mecca ($p \approx 0.2825$, ns), Mt. Meru ($p \approx 0.3956$, ns), and Mt. Kailash ($p \approx 0.3956$, ns) yield no enrichment. Harmonic deviations are computed from Mount Gerizim (35.269°E), which lies within that corridor. The Samaritan *tabbur ha-aretz* ("navel of the earth") tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989) provides independent historical motivation for this choice. Tests using Gerizim reflect properties of the selected corridor, not of any uniquely identified meridian.

Gerizim is not inscribed (UNESCO tentative ref. 5706, DMS E35 16 08); Jerusalem is inscribed and self-excluded when used as anchor.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i to vi), *Natural* (criteria vii to x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (−235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim (35.269°E; not inscribed, §2.2). Jerusalem is inscribed and scored as a regular corpus site. For sensitivity sweeps, Gerizim is added synthetically so each anchor self-excludes symmetrically.

For a site at longitude λ, the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3°$ (folded to $[0, 1.5°]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 16.6$ km for Tier-A+) are equatorial approximations ($\delta \times 111$ km/°). No latitude correction is applied. Kilometer-scale distances should be interpreted as approximate: UNESCO coordinates are site centroids, and precision below ∼1 km is not meaningful.

The geometric null for Tier-A+ is $2 \times 0.15°/1.5° = 10\%$ (assuming uniform longitudes). A Gaussian KDE gives a similar expected A+ count under longitude clumping (101.1 ± 9.6 expected A+). The primary diagnostics use simulation-based nulls (§5.4).

3.3 Statistical Framework

The choice of $(\omega, \phi) = (3°, \text{Gerizim})$ follows from a global parameter sweep. Define $T = \max_{\omega,\phi} S(\omega, \phi)$ where S is the harmonic-deviation enrichment score; T is calibrated by permuting site longitudes and recomputing, correcting for the full search. Global scans over ϕ (36,000 anchors) and unit scans over ω both maximize at 3° / 35°E (§4.1, §5.2; anchor-sweep permutation $p = 0.0753$ †). S tests lattice proximity under a specified frame, not intrinsic periodicity; the two are epistemically distinct (§6.3). Three threshold-free diagnostics below characterize alignment at this fixed frame (see §3.5):

Test 1 Two-component von Mises/uniform mixture fit to the dome sub-corpus ($N = 90$), mean fixed at Gerizim phase, assessed by parametric bootstrap.

Test 2 Spearman rank correlation between harmonic node mean deviation and site count, full corpus ($N = 1011$), permutation null.

Test 3 Phase-rotation null: the reference grid is shifted by a uniform random offset in $[0°, 3°)$; tests whether the Gerizim phase minimizes the full-corpus mean harmonic deviation.

All three point in the same direction (diagnostic p-values in Table 3; Holm-adjusted at $k = 3$). All primary diagnostics operate on the same harmonic-deviation representation and should be interpreted as complementary summaries of a single constructed feature, not as independent evidence. These p-values quantify structure within the constructed frame and are not evidence against spatial randomness in an invariant sense. Tier labels (A++, A+, A) are descriptive vocabulary only and do not enter any diagnostic statistic (§3.4).

3.4 Tier Classification

Sites are assigned to tiers by their angular deviation δ from the nearest 3° harmonic node. The three nested A-windows span the same range of proximities examined in the multi-scale enrichment analysis (§4.5): 4% null (≈6.66 km) through 20% null (≈33 km), covering the full interval over which the four corpora show enrichment.

These windows are not identified as peaks; they are fixed proximity bands used as descriptive vocabulary for summarising harmonic proximity. No primary test statistic depends on a tier cutoff.

Tier A++ $\delta \leq 0.06°$ ($\lesssim 6.66$ km); $P_0 = 4\%$.

Tier A+ $\delta \leq 0.15°$ ($\lesssim 16.6$ km); $P_0 = 10\%$.

Tier A $\delta \leq 0.3°$ ($\lesssim 33$ km); $P_0 = 20\%$.

Tier B Middle zone, between harmonics and inter-harmonic midpoints.

Tier C, C−, C−− Symmetric mirror of A, A+, and A++, measured from the nearest inter-harmonic midpoint.

3.5 Test Populations

Primary threshold-free diagnostics (Tests 1 to 3)

Tests 1 to 3 are described in full in §3.3.

The only additional detail specific to this section is the keyword population for Test 1: 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). The context-validated subset ($N = 83$) restricts to sites where the keyword appears in an architectural context and is reported as a robustness check (Test 5x).

Supporting enrichment tests (Tests 4 to 6)

Tests 4 to 6 count sites falling within a fixed proximity window of the nearest harmonic node and test whether the rate exceeds a geometric null. Three nested windows are reported (Tiers A++, A+, A; null rates 4%, 10%, 20%; widths 0.06°, 0.15°, 0.3° on $[0°, 1.5°)$). No primary diagnostic depends on a tier cutoff (see §3.4).

Test 4 (full-corpus enrichment). Full corpus ($N = 1011$), no keyword filter. Binomial test of A+ rate vs. $p_0 = 10\%$ null.

Test 5 (dome enrichment). Dome keyword subset ($N = 90$). Fisher exact and binomial tests of A+ and A++ rates.

Test 6 (mound evolution, sensitivity). Extends the dome subset with *tumulus, tumuli, barrow, barrows, kofun* ($N = 110$); not in the primary diagnostic family.

Descriptive and post-hoc tests

Test 7 (temporal gradient). Cochran-Armitage trend test on inscription year. Descriptive only.

Test 8 (founding classifier, post-hoc). Keyword classifier; no weight in the primary evidential chain (Appendix C.2).

3.6 Multiple-Comparisons Protocol

Primary diagnostic family ($k = 3$, Holm step-down). Tests 1 to 3 are all threshold-free; they form the primary diagnostic family. The adjustment calibrates these summaries within the selected representation. It is not a confirmatory family-wise test of discovery.

Supporting enrichment tests (Tests 4 and 5). Tests 4 and 5 use a pre-specified proximity window and are not in the primary diagnostic family. Results are in the FDR audit (Table 3; §4.8).

Other. Test 6 is a sensitivity variant of Test 5. Test 7 (temporal gradient) is descriptive. Test 8 is post-hoc. Religion sub-group analyses (§4.4) and diagnostic Tests A to D (§4.3) are not Bonferroni-corrected.

3.7 Simulation Null Models

Three simulation approaches (100,000 trials each), a within-dome bootstrap stability check, and a restricted geographic-concentration null are described with their results in §5.4.

3.8 Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §C.2.

4 Results

Effects are weak in absolute terms (full-corpus A+ rate 13.1% against a 10% null; Spearman $\rho = -0.1076$).

4.1 Spacing Specificity: Is 3° Discriminated from Nearby Intervals?

The 3° spacing is the highest-scoring parameter from the unit sweep (§3.3); metrological context is noted in §2.2. Within the canonical range examined, no nearby random spacing reproduces the observed tier-bin peak: a longitude-preserving permutation of the dome sub-corpus (10000 trials) yields $p = 0.0999$ at A^{+} (∼) and $p = 0.1728$ at A^{++} (ns). Restricting to morphologically unambiguous spherical forms (stupa, tholos, $n = 22$) gives $p = 0.0239$ (*) at the A tier.

A ±0.3% shift from exactly 3° collapses this peak in at least one population, so the result is best read as compatibility with a 3° model rather than precise recovery of a 3° unit (§5.1).

The 1.5° and 6° spacings produce similar tier-bin counts but are mathematical consequences of the 3° structure (shared harmonic nodes). Neither constitutes an independent signal. Candidate intervals at 4° and 8° (§6.2) are similarly arithmetic consequences of the 3° grid. Results support relative optimality of 3° within the tested family. It is a representative optimal scale under this scoring function, not a uniquely identified spacing. Other spacings produce comparable localized structure under alternative parameterizations.

4.2 Phase localization

The 3° model performs better under the tested diagnostics than adjacent alternatives (§4.1); we next ask whether the pattern is uniformly distributed over all possible phases or concentrated at a specific phase. A phase-envelope scan over all origins

Table 1: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.30° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.30° lon)	6.56	4.96	1.32×	
Mann-Whitney U (one-tailed)				0.0025 **
Permutation (10,000 shuffles)			$Z = 3.96$	0.0000 ***
Harmonic-level density (all Tier ≤ B sites per 3° harmonic):				
Harmonics with ≥ 1 A+ site (36)	mean = 25.0 sites			
Harmonics with 0 A+ sites (19)	mean = 5.9 sites		4.24×	< 0.0001 ***

$\phi \in [0, 3°)$ (300 phases at 0.01° resolution) shows a strong preferred phase: full-corpus A+ counts span 83 to 132 (median 100.0), and Gerizim's phase (2.2690° mod 3°) sits at the 100.0$^{\text{th}}$ percentile (132 A+), the 96.7$^{\text{th}}$ for domes (18/90; range 4 to 21). Rayleigh permutation on the 3°-circle: $p_{\text{perm}} = 0.0001$ *** (§5.2). This phase was identified as part of the selected frame (post-selection; §3.3).

4.3 Cluster Asymmetry (Tests 2 and 3, Primary)

Tests 2 and 3 use the full corpus ($N = 1011$), no keyword filter (§3.5), and quantify alignment without any tier cutoff (Table 3). **Test 2** (Spearman rank correlation between harmonic node mean deviation and site count): $\rho = -0.1076$, permutation $p = 0.00065$ (***). Nodes drawing more sites show smaller mean harmonic deviations. **Test 3** (phase-rotation null): the observed full-corpus mean harmonic deviation is smaller than expected when the reference grid is rotated by a uniform random offset in $[0°, 3°)$ ($p_{\text{perm}} = 0.04729$, *). Both point in the same direction (Holm-adjusted diagnostics: $p_{\text{adj}} = 0.002$ **; $p_{\text{adj}} = 0.047$ *).

The tier-based characterisation below provides descriptive context. For each site, count other UNESCO sites within ±0.30° longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.
A+-bearing harmonics host 4.24× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 1). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.25 vs. 6.49 neighbors; 1.43×, $p = 0.0527$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1667) is identical to the unconditional rate (0.1786): in this check, the clustering and unit-sensitivity summaries appear separable.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). Full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns): no spurious global 3° periodicity;

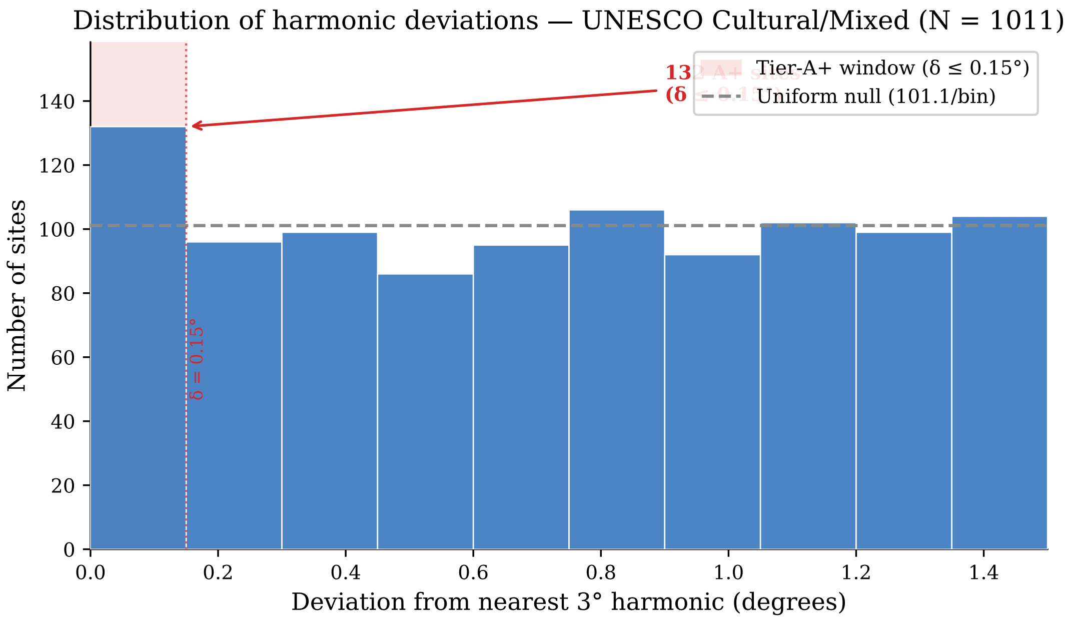

Figure 1: Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.15°$, shaded) above the uniform null (dashed line) illustrates the corpus-level enrichment pattern; Tests 1 and 2 (domed monuments and cluster asymmetry) characterize its morphological and spatial structure.

enrichment is concentrated in A+ sites.

Test B (A+-only Rayleigh, structural check). Among 132 Tier-A+ sites, $R = 0.9842$ ($Z = 23.63$): a structural consequence of the A+ criterion, not independent evidence (§6.1).

Tests C to E (anchor specificity). Circular-shift null: 132 A+ sites vs. null mean 101.05 ($p_{\text{perm}} = 0.0061$, **). Targeted von Mises (full corpus, μ_0 fixed at Gerizim phase): $\hat{w} = 0.0312$ (≈3.1% of sites), $p_{\text{boot}} = 0.0199$ (*). Global anchor sweep (36,000 trial anchors): Levantine corridor $p = 0.0753$ (†).

4.4 Full-Corpus Enrichment (Test 4, Supporting)

We now quantify the excess at the Gerizim-grid phase.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (Figure 1) (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), above the 10% null (binomial $p = 0.0011$; supporting Test 4, not in the primary diagnostic family). The observed enrichment is 1.31×, modest relative to the dome subpopulation (2.00×) and the harmonic cluster asymmetry. The primary threshold-free diagnostics are reported in Table 3; they summarize structure within the selected harmonic representation, not independent confirmation.

Region-stratified audit. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.7978$ (ns), so it does not provide region-stratified evidence against geographic concentration. The pre-2000 cohort remains elevated against the geometric null ($p = 0.0132$ *; §4.7), and is treated as descriptive context.

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum, 2026). A chi-squared test finds no significant A+

rate difference across the five main religion groups ($\chi^2 = 4.753$, df $= 4$, $p = 0.3136$, ns). Both Buddhism (20/71 A-tier, 28.2%; 19/71 C-band, 26.8%; $p = 0.0052$ **) and Hinduism (11/39 A-tier, 28.2%; 10/39 C-band, 25.6%; $p = 0.0324$ *) show a joint A-tier/C-band signal (multivariate hypergeometric). Both persist in tradition-exclusive sites ($p = 0.0123$ *, $p = 0.0862$ †).

4.5 Domed and Spherical Monuments (Test 5, Supporting)

The keyword filter is described in §3.5 ($N = 90$). The dome sub-corpus should be treated as a hypothesis-generating subpopulation: the keyword set was refined iteratively during analysis, and the enrichment should not be read as confirmatory evidence independent of that process. **Test 1** (dome von Mises mixture, Table 3): a two-component von Mises/uniform mixture with the mean fixed at the Gerizim phase yields $\hat{w} = 0.0876$ ($\hat{\kappa} = 252.19$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). Approximately 8.8% of dome sites are concentrated at the Gerizim phase, with an angular spread of $\approx 0.0301°$. The tier-based enrichment below provides complementary descriptive context.

Tier-A++ concentration (supporting result). 11 of 90 dome sites (12.2%, 3.06× the 4% null) fall within Tier-A++ ($\delta \leq 0.06°$, $\lesssim 6.66$ km). This concentration is directionally reproducible across three test statistics: Fisher OR $= 2.71$, $p = 0.0077$ **; permutation $Z = 3.21$, $p = 0.004$ **; bootstrap $p = 0.006$ **. Dome A++ gives consistent results across all three null models and remains directionally robust under full Bonferroni at $k = 44$ (§5.6).

Tier-A+ enrichment (supporting Test 5). 18/90 sites at Tier-A+ (20.0% [12.3, 29.8], 2.00×; Fisher OR $= 1.77$, $p = 0.0346$ *; binomial $p = 0.0032$ **). Tier-A: 29/90 (32.2%, 1.61×; Fisher $p = 0.0175$ *; binomial $p = 0.0043$ **). The χ^2-uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.6586°$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check shows that no single site drives the result: removing any A+ site yields $N = 89$, A+$= 17$, $p = 0.0067$. Leave-2-out (worst case) across all $\binom{18}{2}$ combinations: $p = 0.0133$. Leave-3-out (worst case): $p = 0.0253$; the evolution corpus (Test 6) (§4.6) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{11}{3} = 165$ triples yields $p = 0.0233$, remaining below $\alpha = 0.05$ in this audit (leave-2-out worst case $p = 0.0087$).

Multi-scale combined enrichment. A Fisher combination across 15 log-spaced proximity windows (null rates 0.33% to 40%; canonical tiers A++, A+, A are three of the 15) shows all windows enriched above their null (OR 1.2× to 6.7×, 14 of 15 individually $p < 0.10$); permutation-calibrated $p = 0.00362$ **. No fixed tier cutoff enters this statistic. Applied jointly across all four corpora (UNESCO domes, UNESCO full, Wikidata Q180987 stupas, OWTRAD Silk Road vertices), the maximum joint enrichment falls at approximately 29 km from a harmonic node (1/10th of the ~289 km inter-node spacing at 30°N), with permutation-calibrated $p = 0.01232$ *. Each corpus peaks within this 29 km bound (4 to 29 km across corpora), a scale that emerges from the data rather than any pre-specified threshold.

Context-validated robustness (Test 5x). Sentence-level validation ($N = 83$, 7 rejected): A++ $p = 0.0041$ **; A+ $p = 0.0614$ †. The A++ result strengthens when

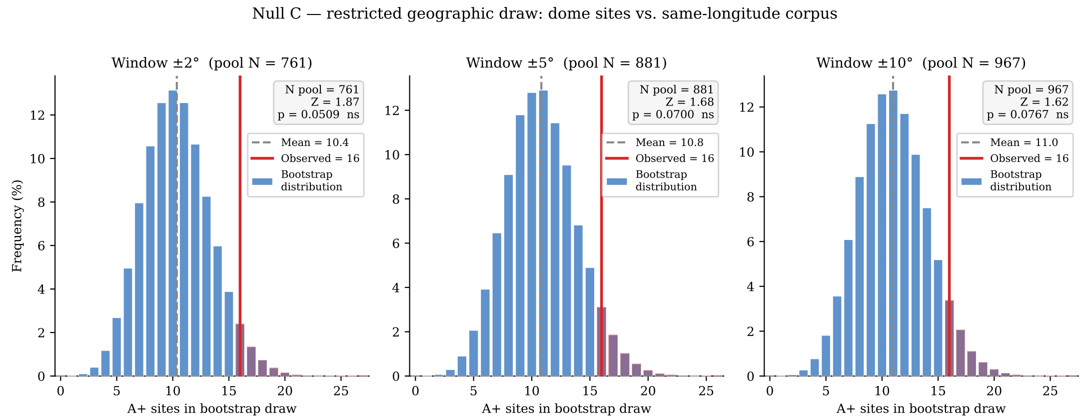

Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed A+ $= 18$) at three window widths. For each window $\pm W°$, all full-corpus sites within $\pm W°$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected A+ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2°$: $p = 0.0257$, *; $\pm 5°$: $p = 0.0393$, *; $\pm 10°$: $p = 0.0442$, *), showing stability across the tested window widths.

false positives are removed; A+ remains directionally elevated under this stricter filter. Keyword-stratified check: stupa/stupas alone ($n = 18$) carries the strongest signal (A++ $p = 0.0057$ **); dome/tholos/spherical without stupa ($n = 65$) remains elevated ($p = 0.0193$ *).

Geographic-concentration checks (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20° to 120°E versus 41% of the full corpus. Two targeted resampling checks evaluate whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap stability check): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 18.00 A+, $p = 0.5429$ (ns). The dome sites' positions are concentrated near harmonic longitudes; this check characterizes the observed dome distribution rather than serving as an external geographic null.

Null C (restricted geographic-concentration null): Pool all full-corpus sites within $\pm W°$ of any dome site and bootstrap the expected A+ count. Dome sites (18 A+, 20.0%) exceed the broader footprint at all three windows tested: $\pm 2°$ ($p = 0.0257$, *); $\pm 5°$ ($p = 0.0393$, *); $\pm 10°$ ($p = 0.0442$, *). Results are stable across the tested window widths.

In the global anchor sweep (§5.2), Gerizim scores 132 A+ (99.7^{th} percentile); Jerusalem scores 128 (97.7^{th}).

4.6 Hemispherical Mound Evolution (Test 6, Sensitivity)

Extending the dome keyword subset of Test 5 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

The combined population ($N = 110$, A+ $= 18.2\%$; Table 2) is consistent with Test 1. The mound stage does not reach significance (8.3%, OR $= 0.60\times$, Fisher $p = 0.8430$, ns). The signal is carried by the stupa (A+ 33.3%, OR $= 3.44\times$, $p = 0.0216$ *) and dome sub-stages (A+ 17.3%, OR $= 1.44\times$, $p = 0.1663$ ns). Leave-one-out checks confirm stability. Leave-3-out (worst case, 1140 triples): $p = 0.0371$.

Table 2: Test 6 enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 4\%$, A+ $P_0 = 10\%$, A $P_0 = 20\%$. Fisher exact tests: stage vs. full corpus background.

		Tier A++ (≤ 6.66 km)		Tier A+ (≤ 16.6 km)		Tier A (≤ 33 km)	
Stage	N	A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound[a]	24	2	8.3% 1.57× (0.3880 ns)	2	8.3% 0.60× (0.8430 ns)	6	25.0% 1.15× (0.4653 ns)
Stupa[b]	18	4	22.2% 5.17× (0.0144 *)	6	33.3% 3.44× (0.0216 *)	7	38.9% 2.22× (0.0874 †)
Dome[c]	75	8	10.7% 2.21× (0.0485 *)	13	17.3% 1.44× (0.1663 ns)	23	30.7% 1.58× (0.0576 †)
Combined	**110**	**13**	**11.8% 2.67× (0.0658 **)**	**20**	**18.2% 1.57× (0.0658 †)**	**35**	**31.8% 1.71× (0.0113 *)**

[a]tumulus/barrow/kofun. [b]stupa/stupas. [c]dome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Table 3: Multiple-comparisons summary. Primary diagnostic family: Tests 1 to 3 (threshold-free, $k = 3$). Supporting enrichment tests (Tests 4 and 5) use a pre-specified $p_0 = 10\%$ proximity window; see §3.5.

Test	N	Raw p	Holm p_{adj}, k=3	Status
Test 1 (dome VM mix.)	90	0.00180^b ($w = 0.0876$, $\kappa = 252.19$)	0.004 **	**Diagnostic**
Test 2 (Spearman)	1011	0.00065^b ($\rho = -0.1076$)	0.002 **	**Diagnostic**
Test 3 (mean dev.)	1011	0.04729^b	0.047 *	**Diagnostic**
Test 4 (full-corpus A+)	1011	0.0011^a	N/A	Supporting
Test 5 (dome A+)	90	0.0032^a (OR = 1.77)	N/A	Supporting
Test 6 (mound evol.)	110	0.0658^a (OR = 1.57)	N/A	Sensitivity
Test 7 (temporal)	1011	0.4184^b	N/A	Descriptive

[a]Binomial or Fisher exact p vs. 10% geometric null (window = 0.15°; convenience choice, sensitivity-audited over 2% to 10% null rates in Supplementary S1); [b]permutation or bootstrap p. Tests 1 to 3 are threshold-free; no tier cutoff enters any primary diagnostic. Adjusted p-values describe calibration within the selected representation, not confirmatory rejection of a global discovery null. Holm (1979) step-down (Holm, 1979).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70° to 110°E). Null A (within-stupa bootstrap): $p = 0.5879$ (ns). Null C (restricted ±5° draw): $p = 0.0556$ (†).

4.7 Temporal Gradient (Test 7, Descriptive)

The Tier-A++ rate in the founding canon (1978 to 1984) is 8.0% (11/138; 1.38× the 4% null, $p = 0.0232$ *), declining monotonically toward the null in later cohorts (2000 to 2025: 4.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.664$, $p = 0.0481$ (*). The five-cohort trend is marginal ($p = 0.0489$ *). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.4184$ ns), consistent with the gradient being specific to the highest-precision tier. This is consistent with UNESCO's founding canon over-representing historically prominent monuments, some of which are also precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate is directionally consistent with the A++ gradient ($Z = -1.760$, $p = 0.0784$ †, OR = 0.722). The declining rate may partly reflect UNESCO's expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

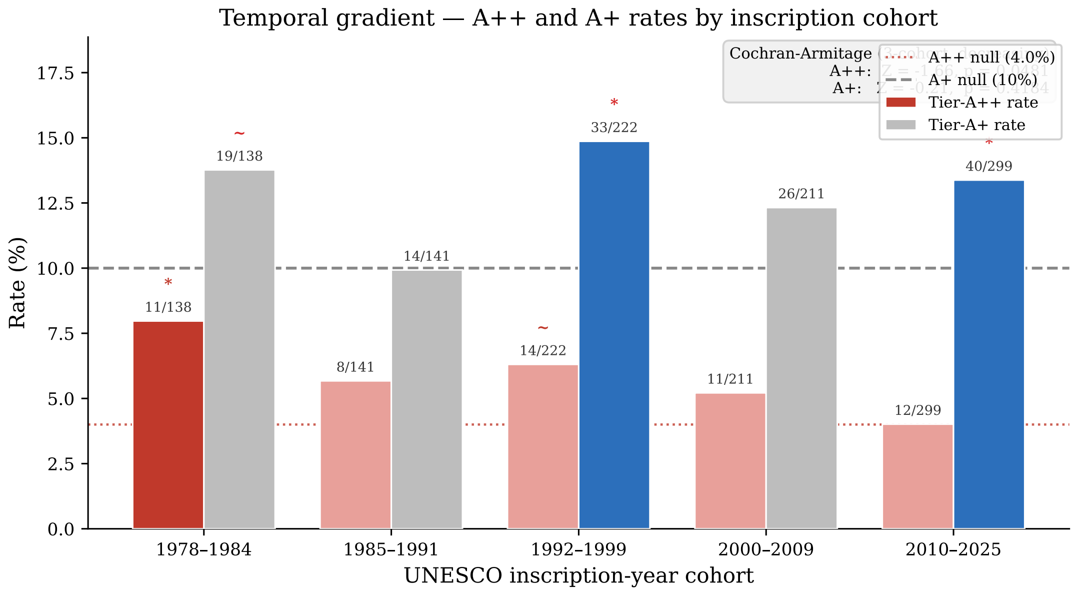

Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978 to 1984) shows an elevated A++ rate (8.0%, $p = 0.0232$ *) that declines monotonically across later cohorts toward the 4% null (dotted line; Cochran-Armitage $Z = -1.664$, $p = 0.0481$ *). The A+ rate (right bars) shows no significant trend ($p = 0.4184$ ns).

4.8 FDR audit and Bonferroni correction

Results are consolidated in Table 3. A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 22 at $q < 0.05$; 6 remain below the full-Bonferroni audit threshold at $k = 44$: dome/spherical A++ ($p = 0.0416$), cluster permutation ($p =< 0.0001$), harmonic Mann-Whitney ($p = 0.0012$), sequential-peak permutation ($p = 0.0279$), and evolution combined A++ ($p = 0.0208$).

5 Robustness Checks

5.1 Unit Sweep and Sensitivity Slope

Unit sweep and sensitivity slope results are in §4.1. Briefly: nearby non-harmonic spacings are non-significant; a ±0.3% shift collapses joint significance; Monte Carlo sweep (§6.2) supports the relative peak; Tests C and D (§4.3) support phase specificity within the selected 3° frame.

5.2 Global Anchor Sweep

The global sweep (36,000 trial anchors, 0.01° resolution) uses the full corpus plus synthetic Gerizim (§3.2), with each anchor self-excluding ($N = 1011$ per trial).

Both anchors place the corridor in the global top 2.3% (36,000 trial anchors, 0 to 360°E; Gerizim: 132 A+, 99.7$^{\text{th}}$ percentile; Jerusalem: 128 A+, 97.7$^{\text{th}}$ percentile). The result is anchor-invariant within the 4.1 km corridor window. Other historical meridians tested here do not show comparable enrichment.

The 35.25°E modulo-3° phase band

Every longitude congruent to 35.25°E modulo 3° (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The flat-top width is a function of the tier threshold chosen: A++ (±0.06°), A+ (±0.15°), A (±0.3°); the number of tied anchors is a methodological artifact, not a physical feature. Mount Gerizim is a fixed landmark at 35.269°E (phase 2.269°, gap 0.019° ≈ 1.8 km). It lies within the A and A+ flat-top bands but not A++.

Formal statistical tests for the 35.25°E modulo-3° phase band

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) shows no global 3° spectral periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0061$, **) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0753$, †, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) identify 3° as the highest-scoring spacing under the template score, *given* the pre-specified Gerizim anchor; neither sweep constitutes anchor-free or period-free discovery.

Corridor and landmark ranking. Of the 13 A+ sites unique to Gerizim and 9 unique to Jerusalem, 119 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #8 among the extended candidate landmarks (97$^{\text{th}}$ percentile).

5.3 Spatial Autocorrelation

A+ harmonics host 4.24× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC (±0.5°, $\rho = 0.187$): DEFF = 1.842, $N_{\text{eff}} = 549$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[10.50\%, 15.43\%]$ excludes the 10% null. Within-group permutation is non-discriminating for alignment vs. geographic concentration; the restricted geographic-concentration null (§4.5) addresses that comparison directly.

5.4 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.060$ †; 1978-1984 canon ($N = 138$) $p = 0.433$ ns; pre-2000 ($N = 501$) $p = 0.490$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.070$ †. (3) KDE-smoothed null: expected 101.1 ± 9.6, observed 132, $Z = 3.23$, $p = 0.001$ **. The KDE null ($p = 0.001$ **) supports dome enrichment; permutation and bootstrap are marginal at the A+ level but stronger for the A++ sub-population (permutation $p = 0.004$ **, bootstrap $p = 0.006$ **). The within-dome bootstrap characterizes the observed concentration near harmonic longitudes, while the restricted geographic-concentration null (Null C; §4.5) indicates that longitude footprint alone is not sufficient to explain the dome enrichment.

5.5 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) in a separately assembled corpus (Tenenbaum, 2026). The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur

(c. 800 CE, 7.23 km, Tier A+) (Miksic, 1990); the World Peace Pagoda at Lumbini achieves A++ precision (∼555 m) under the Gerizim reference frame.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, 1.53× null; binomial $p < 0.001$, ***; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, ∼). Tier-A+ is nominally elevated under the simple binomial count (33/229, 14.4%, binomial $p = 0.021$ *), but the portable signal is strongest at Tier-A because this external-corpus analysis ranks cluster-level occupancy rather than individual-site precision. Its tier emphasis should therefore not be read as mechanically identical to the UNESCO dome scoring. Harmonic nodes with at least one Tier-A site carry 2.56× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.4). The Java/Sumatra node (105° to 115°E, near the 75° harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = 4.74×, $p < 0.001$ ***), rising to 42/54 (77.8%, OR = 18.38×, $p < 0.001$ ***) in the 107° to 112°E window, where A+ is also elevated at 31.5% (OR = 4.57×, $p < 0.001$ ***). The physical heartland, India/Nepal (70° to 90°E) and Myanmar/Cambodia (90° to 105°E), is A-tier depleted (combined OR = 0.21×, $p < 0.001$ ***, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = 4.47×, $p < 0.001$ ***) and the combined heartland 29.2% (OR = 6.00×, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.4). Non-heartland Wikidata stupas (outside 70° to 110°E, $N = 102$) show the same A-enriched, C-depleted profile: 50.0% reach Tier-A (OR = 5.68×, $p < 0.001$ ***) with no C-band enrichment (OR = 0.19×, $p = 1.000$ ns).

5.6 Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 99.7$^{\text{th}}$ percentile; a ±0.3% unit shift collapses joint significance. FDR audit (44 tests) retains 22 at $q < 0.05$; 6 remain below the Bonferroni audit threshold at $k = 44$.

6 Limitations

6.1 Exploratory Status and Multiple-Comparisons Exposure

The analysis is exploratory: spacing, phase, anchor, sub-corpus, and tier thresholds were all examined iteratively on a public dataset for which prospective constraint is not applicable. The Holm correction covers only the three primary diagnostics; p-values are descriptive, not confirmatory. They are not tests of discovery. They quantify how sharply the selected representation describes the data. Key degrees of freedom (3° spacing, Gerizim phase, keyword set, tier thresholds) have external motivation; threshold sensitivity is in Supplementary S1. External corpora are not independent of UNESCO (§7.3).

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe/N. America 50%; Arab States 9%; Asia-Pacific 23%; Latin America

11%; Africa 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0393$, *), which indicates that architectural type may contribute to the enrichment beyond longitude footprint alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacings 0.5° to 15°) gives the highest template score at 3° ($p < 0.001$), with harmonics at 6° and 12° also elevated under the same score. The dome subset yields 1 independent high-scoring spacing ($P = 0.2248$ under the geographic null).

6.3 Representation Dependence

The observed structure depends on the representation. It is recovered under nearest-harmonic-deviation scoring but not by invariant spectral methods (no Rayleigh peak at 3° in any sub-corpus). Three converging diagnostics: (i) a ±0.3% unit perturbation collapses joint significance (§4.1); (ii) multi-scale enrichment is broad, not a sharp spectral line; (iii) the anchor-sweep maximum is marginal ($p = 0.0753$, †) while targeted tests at the identified frame are stronger (§5.2). The pattern appears under the representation, not as intrinsic periodicity. It arises from the interaction between the scoring function and the imposed coordinate frame. A clustered control shows that the unconstrained global maximum over all (ω, ϕ) is typical under realistic dome-site clustering ($p = 0.522$, $N_{\text{sim}} = 2000$) and does not uniquely recover the hypothesised frame. The free-sweep maximizer returns $\omega = 13.75°$, a spacing with no metrological or waypoint interpretation, confirming that the unconstrained maximum reflects spatial clustering rather than any meaningful periodicity. This outcome is expected: the enrichment statistic uses a fixed absolute threshold (0.15°), so large spacings receive an artificially low null rate and therefore inflate the enrichment ratio regardless of any real structure. The free sweep is uninformative precisely because it is not scale-invariant; the metrologically motivated prior $\omega = 3°$ is tested conditionally, not recovered by unconstrained search.

6.4 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate 9.7%, below the 10% null ($p = 0.5149$ ns, one-sided depletion).

6.5 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids (unsystematic noise, no directional bias).

Metrological evidence. The 3° template was initially motivated as a base-10 subdivision of the 30° bēru (Hunger & Steele, 2019; Powell, 1987). No ancient surveying instrument calibrated to such sub-units has been recovered, and the metrological parallel is treated as heuristic motivation, not evidence. All thresholds are degree-denominated.

7 Discussion

7.1 Template Alignment vs. Intrinsic Periodicity

The present results are best treated as *template alignment* (§6.3): a reproducible, phase-localized pattern appears under the representation but is not invariant and does not imply a periodic generative model. The analysis does not demonstrate an underlying spatial law or intentional placement. It shows that coordinate-aligned statistics can produce reproducible deviations in clustered datasets. Interpretation should focus on properties of the representation, not the existence of a latent periodic structure.

7.2 Alternative Explanations

(1) Independent convergence. Hemispherical structures recur across cultures (Aveni, 2001, 2003), but convergence predicts no longitude specificity.

(2) Geographic concentration. Null C: other monuments from the same longitude footprint are less enriched than the dome subset ($p = 0.0393$, *); these checks reduce, but do not eliminate, the possibility that enrichment partly reflects corridor concentration.

(3) Random coincidence. Simulation nulls (KDE, permutation, bootstrap) support dome enrichment within the selected representation, most strongly at A++ (permutation $p = 0.004$ **; full results in §5.4). Tests C and D support phase specificity within the selected frame (§4.3).

7.3 Cross-Corpus Consistency: OWTRAD and Wikidata

Two additional corpora (the Wikidata Q180987 stupa inventory and the OWTRAD route network (Ciolek, 2004)) are analyzed under the same harmonic-deviation representation. These datasets are not statistically independent of the UNESCO corpus, as they share geographic and cultural structure. Their role here is not replication, but to assess whether the observed alignment behavior is stable across related datasets under a common representation.

OWTRAD route network. The OWTRAD route network contains 1946 route segments and 1674 unique endpoints. Under the same representation, the degree-weighted endpoint distribution shows A++ enrichment (211 of 3892 endpoint positions, $z = 4.53$, $p < 0.0001$ ***), while the deduplicated endpoint distribution is weaker (84/1674, $p = 0.0224$ *). A phase-shift test gives $p = 0.0202$ *, and a cluster-asymmetry check gives $p = 0.0007$ ***. These results are treated as shared-representation consistency, not independent replication. The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

8 Conclusion

This paper reports an exploratory pattern: the UNESCO Cultural and Mixed corpus shows a consistent template-alignment excess near the ∼35°E corridor under multiple diagnostic checks (Table 3; $p_{\text{adj}} = 0.004, 0.002, 0.047$), with small effect sizes and corridor-level rather than point localization. These p-values describe within-frame consistency, not confirmatory evidence.

A multi-scale combined test reduces dependence on any single tier choice (dome subset $p = 0.00362$ **; joint enrichment peaks near 29 km, $p = 0.01232$ *). The Wikidata stupa corpus and OWTRAD route network show similar enrichment patterns under the same reference frame (§7.3).

In short, a harmonic-alignment statistic identifies a weak but reproducible pattern under a specific coordinate representation. This pattern is not invariant, does not support a periodic generative model, and is consistent with the interaction

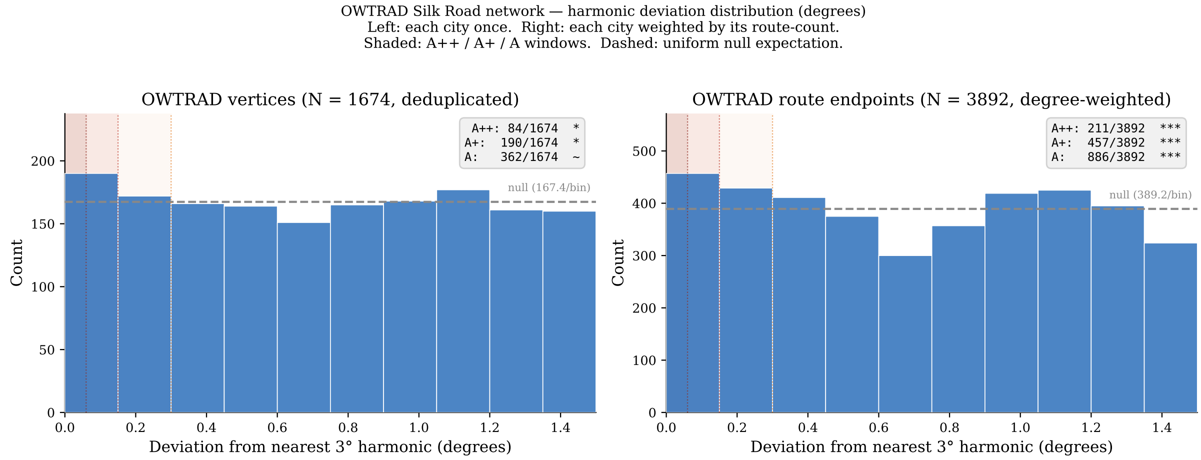

Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3892 degree-weighted route endpoint positions (right). The degree-weighted distribution shows A++ enrichment ($z = 4.53$, $p < 0.0001$ ***); the deduplicated endpoint distribution is weaker ($z = 2.13$, $p = 0.0224$ *).

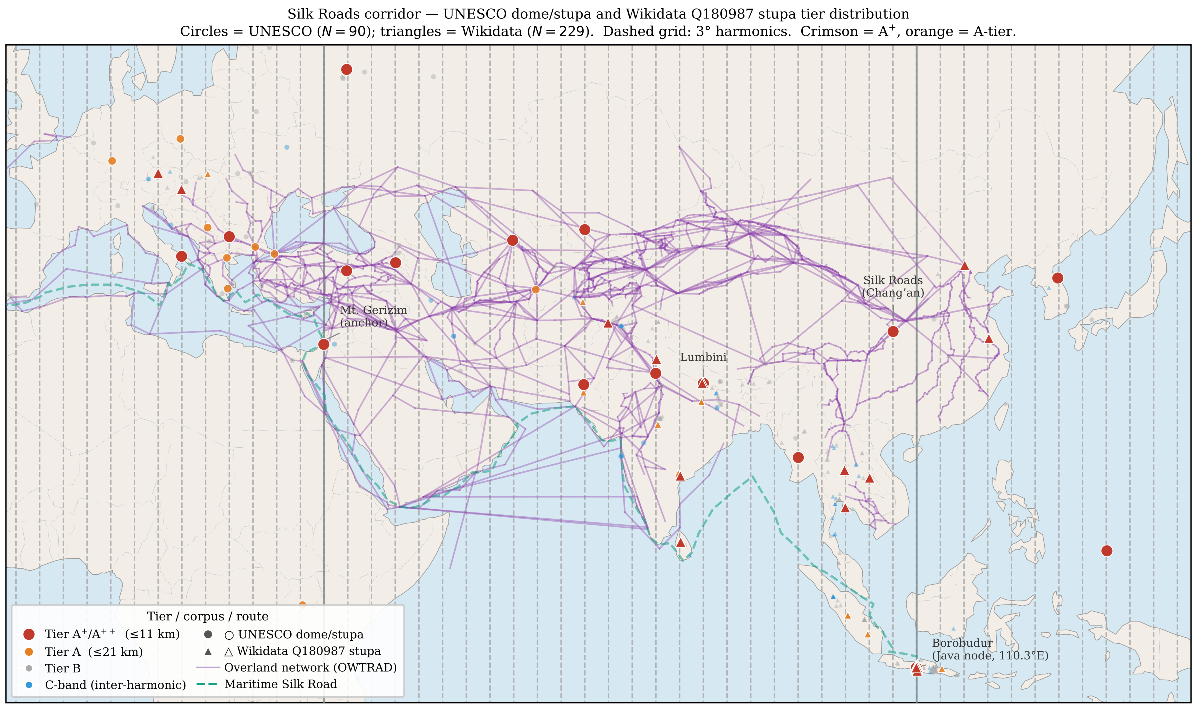

Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (`supplementary/interactive_map/silkroad_interactive.html`).

between geographic clustering and the imposed harmonic frame. Its primary contribution is methodological. It shows how representation choice can induce stable but non-invariant spatial patterns in culturally clustered datasets. Independent re-examination on corpora without UNESCO's Eurasian sampling structure is the appropriate next step.

A Von Mises Mixture Characterisation of External Corpora

A von Mises + uniform mixture model $f(\theta|\kappa) = w \cdot \mathrm{VM}(0, \kappa) + (1-w) \cdot \mathrm{Uniform}(0, 2\pi)$ was fitted to the harmonic phase deviations ($\theta \in [0, \pi]$, $\theta = 0$ on-harmonic) of two external corpora using maximum likelihood with a grid of starting values.

OWTRAD Silk Road nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = \kappa^{-1/2} = 0.1543°$.

Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922°$.

N-weighted pooled $\hat{\sigma} = 0.1588°$; κ ratio $= 1.6$ across corpora.

These values characterize external-corpus clustering and are not used to derive tier boundaries; agreement with tier widths is descriptive context.

B Tier-Threshold Sensitivity Audit

A log-scale null-rate sweep across 73 thresholds supports the interpretation that enrichment is not an artifact of the specific tier width: 57 of 73 thresholds reach $p < 0.05$ for the dome subset; dome OR ranges 1.53×–3.40× across the three canonical tiers (A, A+, A++). Within ±1 log-decade of the A+ threshold, 27 of 28 thresholds are significant. Full results in `supplementary/audit/tier_sensitivity_audit.txt`.

C Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`; Tenenbaum 2026); regenerate all with `bash tools/update_all_audits.sh`.

C.1 Dome/Spherical Monument Filter (Tests 1, 5, 5x, 6x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in the Test 1 dome mixture and Test 5 tier-enrichment analyses without context gating.

For Test 5x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres.

Test 6 adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 6x applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.5 and §4.6.

C.2 Founding-Origin Classifier (Post-Hoc)

Results. The classifier finds 31.8% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.19, $p = 0.225$ ns). The A++ sub-population (56 sites) shows a stronger signal: 42.9% carry a founding keyword (OR = 1.94, $p = 0.014$ *), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

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Data Availability Statement

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; `https://doi.org/10.5281/zenodo.19938283`); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`), with each file's role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

Acknowledgements

The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted independently and without external institutional affiliation.

Conflict of Interest

The author declares no conflict of interest.

Software and AI Disclosure

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.